

Real-Time Transfer Function Editor for Direct Volume Rendering in Mixed Reality

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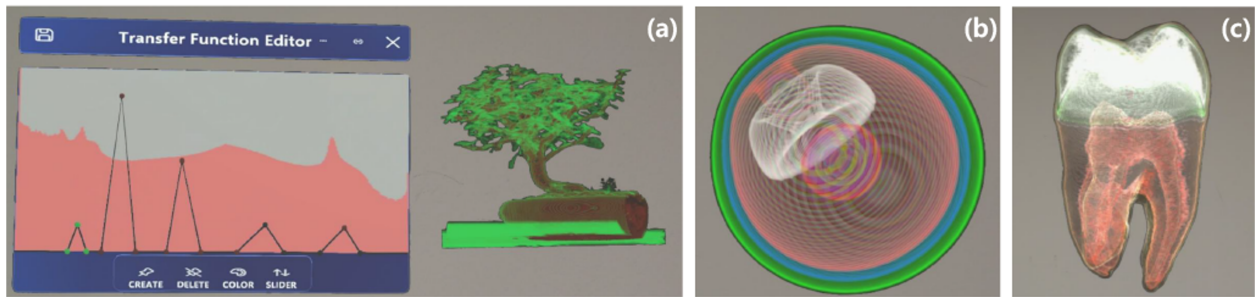


Figure 1: Examples of (a) our TF editor with bonsai, (b) nucleon, and (c) tooth rendering highlighting internal structures.

Abstract

Direct Volume Rendering (DVR) effectively visualizes volumetric data, allowing focus on specific regions of interest (ROIs) while retaining surrounding context. In mixed reality (MR), DVR offers an immersive and intuitive way to interact with volumetric data. However, the ability to adjust transfer functions (TF), which control the appearance of DVR, has been limited in MR settings, often relying on predefined TFs that restrict real-time customization. In this paper, we introduce a novel system designed to enhance DVR within MR environments by providing an interactive TF editor. This system allows users to intuitively adjust the optical properties of volumetric data through a user-friendly interface. Key features include node creation and deletion for defining ROIs, a color palette and opacity control for appearance settings. Immediate visual feedback is achieved by converting the user-defined 1D TF into a 2D texture that is applied in real-time using a DVR Unity shader.

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1 Introduction

Direct volume rendering (DVR) is a three-dimensional visualization technique for volumetric data that has proven effective in various application areas [Fuchs and Hauser 2009; Kaufman et al. 1994]. A key feature of DVR is its ability to visually highlight specific regions within a volume while still providing near context. This allows users to focus on areas of interest (ROIs) and facilitates the effective analysis and interpretation of data. Transfer functions (TF) are essential in the DVR process, as they allow users to interactively determine which regions are displayed and how these ROIs are represented. This is accomplished by assigning appropriate optical properties, such as opacity and color, to all voxels within the volume.

Mixed reality (MR) combines real-world environments with virtual objects, creating an interactive and immersive workspace where the physical and virtual worlds coexist. This environment demonstrates the potential of DVR by allowing users to visualize and interact with volumetric data in a more intuitive and engaging manner. However, the adjustment of TF, which affects DVR appearance, has been limited within MR environments. Conventionally, users have relied on predefined TFs, which significantly restricts their ability to interactively manipulate visualizations in real-time to access desired information. This limitation hinders the potential of MR to fully leverage DVR capabilities for enhanced user experience and data interpretation.

In this paper, we introduce an interactive TF editor system designed to enhance DVR within MR environments (see Figure 1).

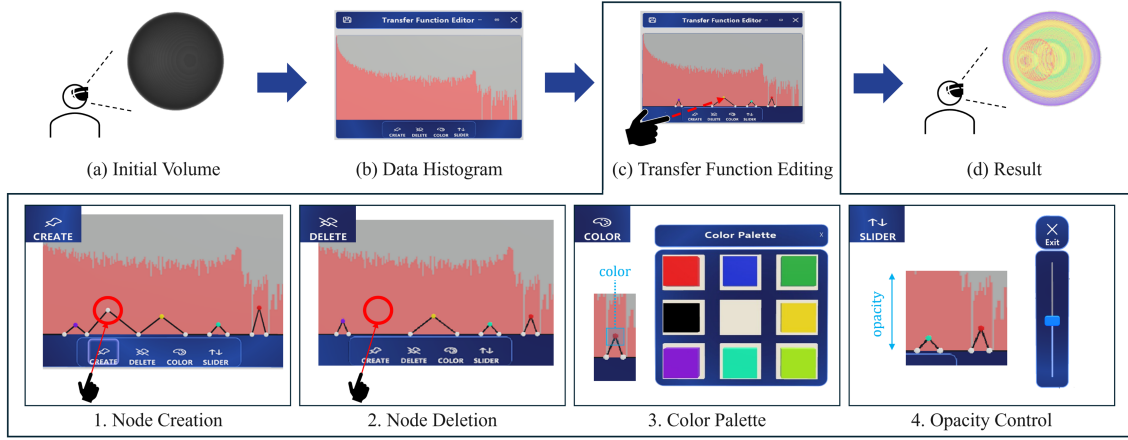


Figure 2: Overview of the transfer function editing process and functionalities.

This system allows users to intuitively adjust the optical properties of volumetric data, providing a more immersive and user-centered approach to data visualization. By enabling real-time manipulation of transfer functions through an intuitive interface, users can interactively define ROIs and apply precise color and opacity settings.

2 IMPLEMENTATION

2.1 Data histogram representation

Histogram of volume provides a fundamental insight into the volumetric data, allowing users to understand the information of volume through distribution patterns simply (see Figure 2-b). For example, multiple clustered peaks may indicate several important regions within the data, while steep slopes could suggest boundaries or structures with large density differences. However, histogram can sometimes present challenges such as unbalanced data value, leading to misleading interpretations. To mitigate these issues and enhance user comprehension, we have normalized the data distribution using the following equation:

$$H_i = \frac{\ln(h_i + 1)}{\ln(h_{max} + 1)} \cdot (r_e - r_s) + r_s \quad (1)$$

where H_i represents the normalized histogram value for i -th bin, h_i is the original histogram count in that bin, and h_{max} is the maximum count across all bins. The term r_e and r_s correspond to the upper and lower bounds of the desired range, respectively. The equation 1 logarithmically scales the histogram values to enhance the visibility of less frequent bins while maintaining the relative magnitude, and then maps these scaled values to the specified range $[r_e, r_s]$.

2.2 Transfer function editor

The TF editor is based on a 1D TF mapping system and provides users with four core functionalities: node creation, node deletion, color palette, and opacity control (see Figure 2-c). These functions are controlled through hand interactions using a Ray, allowing users to interact with the volume easily and intuitively. The "node creation" and "node deletion" functions focused on determining the ROI of the volumetric data that the user wants to visualize. By placing nodes on the histogram, users can effectively map out specific

intensity ranges, enabling them to define particular structures or regions within the data.

The "color palette" function simplifies selecting precise RGB values in MR environment, where selecting precise colors in the color space can be difficult. When a node is selected, a color palette appears, enabling the user to choose the exact RGB values that best represent the data segment. Additionally, the "opacity control" is a crucial addition in MR, where adjusting the opacity of a node without changing its position can be challenging. This feature provides separate controllers for each node, allowing users to finely adjust the opacity values to control the visibility of specific regions within the volume, without affecting the node's position.

2.3 Texture application

Our system provides immediate visual feedback by converting the user-generated 1D TF into a 2D texture on per-frame basis. The 2D texture is set to size of the data range of target volume (width) by 1 (height) and is dynamically updated. Color and opacity values between TF nodes are determined through linear interpolation to ensure smooth transitions. It is then applied to the volumetric data through a DVR Unity shader.

3 CONCLUSION AND FUTURE WORK

In this work, we present a system that assists users in creating and refining volumetric data visualizations within MR environments. The system enables intuitive and dynamic adjustments of optical properties, allowing users to focus on specific ROI through real-time manipulation of color and opacity adjustments. For future work, we plan to explore more advanced interaction techniques, such as voice commands and gesture recognition, to further enhance the usability of the TF editor in MR environment.

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